

TUM and Network Flow

Recall: given a directed graph, incidence matrix is signed. $\{-1, 0, +1\}$.

$$\underline{\delta^-}, \underline{\delta^+}$$

Flow conservation:

$$\boxed{\sum_{e \in \delta^-(v)} x_e = \sum_{e \in \delta^+(v)} x_e}$$

$\forall v$ w/o supply/demand

- so for max flow, any node $v \neq s, t$.

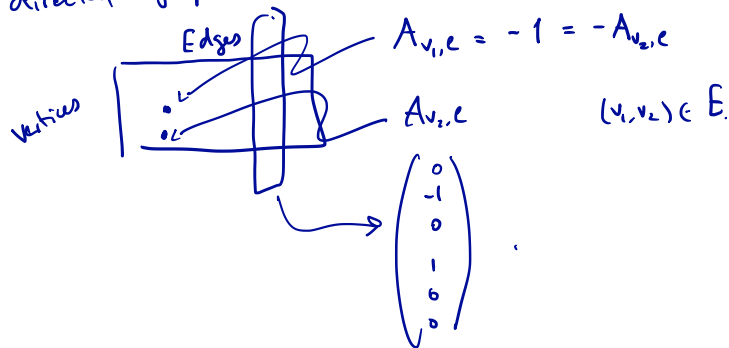
Max
Flow

$$\text{max: } \sum_{e \in \delta^+(s)} x_e$$

$$\text{st: Flow conservation } \forall v \neq s, t \\ x_e \geq 0$$

The Flow Matrix is TU

Let A be signed ^{incidence} matrix of a directed graph.



Thm: A is TU.

Exercise. Hint - see proof that A for bipartite graph is TU.

Max Flow is Integral

If capacities are integral, then
the ~~the~~ max flow solution is integral.

pf.

Follows from above.

Also using if A is TU

$\Rightarrow$ any submatrix is TU.

Max Flow / Min Cut

Recall a cut is a collection of edges s.t. w/o those edges there is no $s-t$ path.

Equiv: $W, \bar{W}$ $s \in W, t \in W^c$

$$\max: w^T x$$

$$\text{s.t. } A'x = 0 \quad z$$

$$x \geq 0$$

$$x \leq c$$

$$y$$

$$\min: 0^T z + c^T y$$

$$\text{s.t. } (A')^T z + Iy \geq w$$

$$y \geq 0$$

Ex 1: primal & dual have integral sol'n $\forall b, c$ integral.

Ex 2: conclude $\max \text{ flow} = \min \text{ cut.}$

Characterizations of TU

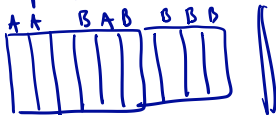
Thm: For $A_{ij} \in \{-1, 0, +1\}$, T.F.A.E.

1. Each square submatrix of A has $\det 0, \pm 1$.

2. $P = \{x: x \geq 0, Ax \leq b\}$ has integral vertices
 $\forall b \in \mathbb{Z}^m$.

3. $\forall$ integral a, b, c, d , $P = \{x: c \leq x \leq d, a \leq Ax \leq b\}$
has integral vertices.

4. Each collection of columns of A can be split into
2 parts, so that sum of cols in one part, minus
sum of cols in other part, is a vector with entries
in $0, 1, -1$.



Characterizations of TU — cont'd

5. Each non-singular sq. submatrix has a row with an odd # of non-zero components.
6. The sum of entries in any sq. submatrix with even # rows & col sums, is divisible by 4.
7. No sq. submatrix of A has $\det \pm 2$.

Not All TUM Are Network Matrices

Ex:
$$\begin{bmatrix} 1 & -1 & 0 & 0 & -1 \\ -1 & 1 & -1 & 0 & 0 \\ 0 & -1 & 1 & -1 & 0 \\ 0 & 0 & -1 & 1 & -1 \\ -1 & 0 & 0 & -1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

Operations Preserving TU — 1/2

- i. Permute rows / columns.
- ii. Take transpose
- iii. Multiply row / column by -1 .

iv. Pivoting:
$$\begin{bmatrix} \epsilon & c \\ b & D \end{bmatrix} \rightarrow \begin{bmatrix} -\epsilon & \epsilon c \\ \epsilon b & D - \epsilon b c \end{bmatrix}$$

$\epsilon = \pm 1$, b, c vectors

- v. Add all-zero row or column, or add a row or column with single non-zero ± 1 .
- vi. repeat row / column.

Operations Preserving TU — 2/2

$$\begin{array}{l}
 \text{vii.} \quad \underline{A} \oplus_1 \underline{B} \triangleq \begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix} \\
 \text{viii.} \quad \underline{[A \ a]} \oplus_2 \underline{\begin{bmatrix} b \\ B \end{bmatrix}} \triangleq \begin{bmatrix} A & ab \\ 0 & B \end{bmatrix} \\
 \text{ix.} \quad \underline{\begin{bmatrix} A & a & a \\ c & 0 & 1 \end{bmatrix}} \oplus_3 \underline{\begin{bmatrix} 1 & 0 & b \\ d & d & B \end{bmatrix}} \triangleq \begin{bmatrix} A & ab \\ ac & B \end{bmatrix}
 \end{array}$$

Seymour's Decomposition Theorem

Thm: A is TU iff A

arises from network matrices

and 2 5×5 matrices given a few slides ago

by applying the above operations, i - ix

Operations vii, viii, ix are applied only
when $A \dot{\cup} B$ are large enough:

$\# \text{ rows} + \# \text{ cols} \geq 4$.

TUM

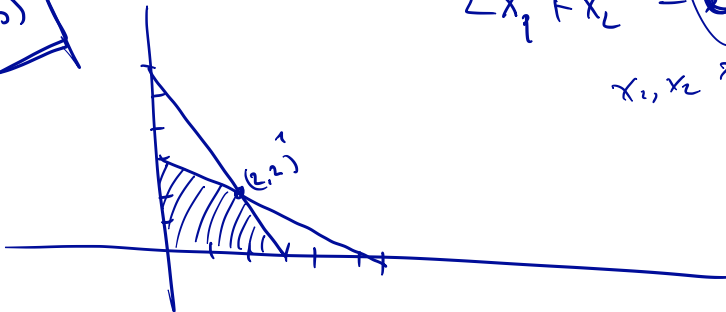
$\{Ax \leq b\}$ integral $\forall b$.

(A)

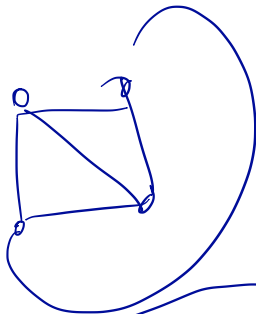
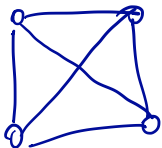
(A, b)

$$\begin{aligned} x_1 + 2x_2 &\leq 6 \\ 2x_1 + x_2 &\leq 6 \end{aligned} \quad \begin{matrix} 5 \\ 5 \end{matrix}$$

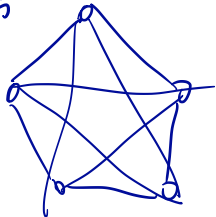
$x_1, x_2 \geq 0$



$\text{Thm } 1.7$
 12/11/17
 G
~~12/11/17~~



K_5



X

$K_{5,5}$

